Prasanna Kumar Routray

Curriculum Vitae

ITEE-CSE
University of Oulu, Finland
(+91) 9438555358, (+358) 0449502427
prasanna.routray97@gmail.com
My Webpage
Github *Linkedin* *Skype*

I am a postdoctoral researcher in virtual reality, neurocognitive analysis, and human-computer interaction. My current work in Professor Steven LaValle's group at the University of Oulu uses VR and ERP/EEG to study motion sickness, sensory conflict, postural instability, and human-subject responses. My doctoral research focused on bioinspired tactile sensing, whisker-like sensors, texture classification, and haptic sensing. I aim to develop sensory interfaces that are technically robust, perceptually meaningful, and comfortable for users.

Research Interests

Haptics; human-subject studies; bioinspired sensor design; robotics; ERP/EEG; virtual reality; tactile perception; human phenotype; and phenotype-informed stimulus interface design.

Education

2025–Present **Postdoctoral Researcher, Faculty of Information Technology and Electrical Engineering**, *University of Oulu*, Finland.
Research areas: Neurocognitive studies, virtual reality, ERP/EEG, motion sickness, sensory conflict, postural instability, and user studies.

2020–2025 **PhD, Applied Mechanics & Biomedical Engineering**, *Indian Institute of Technology Madras*, India.
Bioinspired sensor design, Whisker sensor, Sensor analysis, sensor response separation, Virtual reality, Haptics, User study

2014–2016 **Master of Engineering, Mechatronics**, *Universidade de Aveiro*, Aveiro, Portugal, Industrial robotics, Industrial vision, Digital control, Depth estimation, Point-cloud segmentation, Robot Control.

2009–2013 **Bachelor of Engineering, Electrical Engineering**, *Government College of Engineering, Keonjhar*, Odisha(*Affiliated to* Biju Patnaik University of Technology, Rourkela, Odisha, India), Smart home, Moisture controlled irrigation.

Research Experience

University of Oulu, Finland

June 2025 – present ***Neurocognitive Analysis of Motion Sickness in Virtual Reality***.
Study of Motion Sickness, Sensory Conflict, and Postural Instability Using VR and ERP.
Advisor : **Dr. Steven M. LaValle, Dr. Even G. Center**

Indian Institute of Technology Madras, India

Feb 2023 – Feb 2025 ***Preimage-Based Framework for Whisker Sensing and Tactile Localization***.
Developed a preimage-based framework for whisker sensing and tactile localization.
Advisor : **Dr. Manivannan M., Dr. Steven M. LaValle, Dr. Basak Sakcak**

Aug 2022 – Feb 2024 ***Whisker Sensor Response Separation***.
Separating Intrinsic and Extrinsic Responses of Whisker Sensors Using Accelerometer.
Advisor : **Dr. Manivannan M., Dr. Pauline Pounds, Dr. Basak Sakcak**

Aug 2021 – Jun 2024 ***Design and Evaluation of Bioinspired Whisker-Like Touch Sensors***.
Designed and evaluated bioinspired whisker-like tactile sensors.

Advisor : **Dr. Manivannan M., Dr. Pauline Pounds**

Aug 2022 – Feb 2024 ***Thin Film Haptic Sensor***.
Developed and evaluated Subblescope, a thin-film haptic sensor.

Advisor : **Dr. Manivannan M.**

Jun 2021 – Aug 2022 ***Surface Texture Analysis Using Whisker Sensor***.
Classification of Multi-dimensional Texture Using a Whisker Sensor and Accelerometer.

Advisor : **Dr. Manivannan M., Dr. Pauline Pounds**

Universidade de Aveiro, Aveiro, Portugal

July 2015 – May 2016 ***Articulated Robot Catching a Flying ball Using Kinect Sensor***.
Development of a system to aid the articulated robot to catch a flying ball using Kinect like depth sensor.

Advisor : **Dr. Filipe Miguel Silva, Dr. Paulo Dias**

Feb 2015 – Jun 2015 ***Upper Arm Movement Imitation Using Kinect Sensor***.
Development of upper-body movement imitation system using V-REP, MATLAB and Kinect Sensor.

Advisor : **Dr. Filipe Miguel Silva**

Publications

Articles

Refereed Journal Articles

Prasanna K Routray, Debadutta Subudhi, Basak Sakcak, Steven M LaValle, Pauline Pounds, and M Manivannan. Separating intrinsic and extrinsic responses of whisker sensors using accelerometer. *IEEE Sensors Journal*, 2024.

Debadutta Subudhi, Prasanna K Routray, and Manivannan Muniyandi. Subblescope: Novel thin film haptic sensing using a single bubble approach. *IEEE Sensors Journal*, 2024.

Asif Ahmed Sardar, Aravinda S. Rao, Tansu Alpcan, Goutam Das, Prasanna Routray, Manivannan M, and Marimuthu Palaniswami. Application-aware real-time network resource allocation for industry 5.0. *IEEE Transactions on Cognitive Communications and Networking*, pages 1–1, 2024.

Journal Articles Under Review

Prasanna Routray, Basak Sakcak, Steven M. LaValle, and Manivannan M., A Preimage–Based Framework for Whisker Sensing and Robot Localization, In *Springer Autonomous Robots*.

In Conference Proceedings

Refereed Conference Publications

Prasanna Kumar Routray, Debadutta Subudhi, Pauline Pounds, et al. Whisker-like sensors: Effective design. In *2025 13th International Conference on Control, Mechatronics and Automation (ICCMA)*, pages 435–440. IEEE, 2025.

Prasanna Kumar Routray, Aditya Sanjiv Kanade, Kshitij Tiwari, Pauline Pounds, and Manivannan Muniyandi. Towards multidimensional textural perception and classification through whisker. In *2022 IEEE International Symposium on Robotic and Sensors Environments (ROSE)*, pages 1–7. IEEE, 2022.

Kshitij Tiwari, Basak Sakcak, Prasanna Routray, M Manivannan, and Steven M LaValle. Visibility-inspired models of touch sensors for navigation. In *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 13151–13158. IEEE, 2022.

Pranav Deshpande, Prasanna Kumar Routray, Soumya Dutta, and M Manivannan. Simple fabrication process for high-sensitive flexible capacitive force sensor using pdms. In *2023 IEEE SENSORS*, pages 1–4. IEEE, 2023.

Rishabh Shukla, Prasanna Kumar Routray, Debadutta Subudhi, and M Manivannan. Whiskered contact-based non-intrusive vibrometer. In *2022 10th International Conference on Control, Mechatronics and Automation (ICCMA)*, pages 161–167. IEEE, 2022.

Rishabh Shukla, Prasanna Kumar Routray, Kshitij Tiwari, Steven M LaValle, and M Manivannan. Monofilament whisker-based mobile robot navigation. In *2021 IEEE World Haptics Conference (WHC)*, pages 1150–1150. IEEE, 2021.

Teaching Experience

Spring 2022: **AM5019: Haptics in Biomedical Engineering**, IIT Madras.
Fall 2023: **AM5011: Virtual Reality Engineering**, IIT Madras.
Spring 2023: **AM5019: Haptics in Biomedical Engineering**, IIT Madras.
Fall 2024: **AM5011: Virtual Reality Engineering**, IIT Madras, on-Campus.
Fall 2024: **Mastering VR : Fundamentals to Practice**, IIT Madras-Finland, NPTEL.
Advisor: **Dr. Manivannan M., Dr. Anna LaValle**

New Courses Proposed for Development

Course 1: Fundamentals of Sensing and Tracking: From Sensing Principles to Sensor Fusion
Course 2: From Sensing to Perception: Human-Centered Experimental Methods

Technical Skills

Programming	Python, MATLAB, C/C++, Julia, Visual Basic
VR and HCI	Virtual reality experiment design, human-subject studies, user-study protocol development, haptic interaction, motion-sickness experiments, and behavioral data collection
Neurocognitive Methods	ERP/EEG experiment design, event-marker synchronization, EEG preprocessing, artifact rejection, and analysis of neurocognitive responses
Haptics and Sensors	Bioinspired tactile sensing, whisker-like sensors, flexible force sensors, haptic sensing, sensor calibration, signal processing, and texture classification
Robotics and Embedded Systems	ROS, RTOS, STM32, Raspberry Pi, wheeled robots, collaborative manipulators, depth cameras, and Kinect-based tracking
Design and Simulation	SolidWorks, KiCad, e-Plan, COMSOL

Industry Experience

Industrial Consultancy and Sponsored Research, IIT Madras, India

Jul 2019–Dec 2019 ***CPR Performance Evaluation in VR***.
Depth Sensor Based CPR Performance Evaluation in VR

Supervisor : **Dr. Manivannan M.**

Atena - Automação Industrial, Lda, Palhaça, Portugal

Aug 2016–Jun 2017 ***Assembly line for diesel engine oil-pumps***.
Automatic assembly of diesel engine oil pumps for Renault passenger vehicle

Aug 2016–Jun 2017 ***Monitoring of energy consumption in industries***.
Monitoring of energy consumption in industry using DMG800 and Visual basic through TCP/IP communication

Supervisor : **Abilio Manuel Ribeiro Borges**

Freelance

Jan 2018–Mar 2019 ***Fitness Training Scoring Mechanism Using Depth Sensors***.
Tracking of medicine balls using depth cameras for real-time performance evaluation during gym session and strength training

Fellowships & Awards

2025–present **Postdoctoral Researcher funded by a European Research Council Grant**, *University of Oulu*, Finland.

2020–2024 ***Research Assistant Fellowship*** of Ministry of Human Resources Development (MHRD), Government of India, as a PhD research scholar in Indian Institute of Technology Madras.

Professional Membership

2023-present **IEEE Member**.

2021-2023 **IEEE student member**, *Madras Chapter*.

Position of Responsibility

June 22-27, 2025 **Workshop Organizer, HCI International 2025**, Gothenburg, Sweden.

Nov 16-17, 2024 **Chief Coordinator, XR Summit 2024**, IIT Madras, Chennai.

Sept 2024 **Coordinator, IViL - IIT For Villages**, IIT Madras, Chennai.

Jul 2024 **Collaborative robot procurement (USD 30000.00)**, IIT Madras, Chennai.

Sept 13-14, 2022 **Chief Coordinator, Wellness Workshop**, IIT Madras, Chennai.

Referees

Dr. Manivannan M.
Professor, Department of
Applied Mechanics & Biomedical Engineering
Indian Institute of Technology Madras, India
☎ +(91) 44-2257-4064
✉ mani@iitm.ac.in

Dr. Steven M. LaValle
Professor, Faculty of
Information Technology & Electrical Engineering
University of Oulu, Finland
☎ +(358) 46-921-7482
✉ steven.LaValle@oulu.fi

Dr. Pauline Pounds
Professor, School of
Electrical Engineering & Computer Science
The University of Queensland, Australia
☎ +(61) 73-365-2883
✉ pauline.pounds@uq.edu.au

Dr. Başak Sakçak
Assistant Professor
Advanced Computing Sciences
Maastricht University, The Netherlands
☎ +(358) 41-752-6523
✉ basak.sakcak@maastrichtuniversity.nl